

Question 1**(Begin a new page)****Marks**

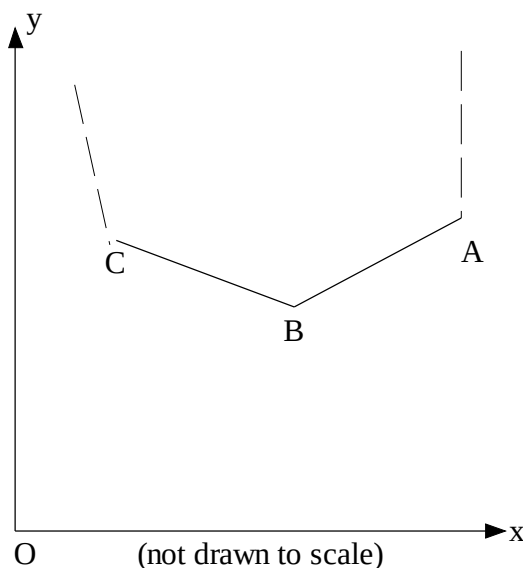
- (a) (i) Show that $\frac{1}{e^x - 1} = \frac{e^{-x}}{1 - e^{-x}}$. **1**
- (ii) Hence find $\int \frac{1}{e^x - 1} dx$. **1**
- (b) Find $\int \frac{x}{\sqrt{1+x}} dx$ using the substitution $u = 1 + x$. **2**
- (c) Use partial fractions to find $\int \frac{x^2 + x - 4}{(2x + 1)(x^2 + 4)} dx$ **3**
- (d) (i) Use integration by parts to find $\int \tan^{-1} x \, dx$. **2**
- (ii) Hence find $\int x \tan^{-1}(x^2) \, dx$. **2**
- (e) (i) Find $\int \frac{dx}{x^2 + 2x + 10}$ **2**
- (ii) Use the result $\int_a^b f(x) \, dx = \int_a^b f(a + b - x) \, dx$ to evaluate the integral $\int_0^\pi x \cos 2x \, dx$. **2**

Question 2**(Begin a new page)**

- (a) Given $w = 2 + 5i$, $z = 4 - 3i$ and $u = a + bi$, evaluate:
- (i) $w + \bar{z}$ **1**
- (ii) $|w + \bar{z}|$ **1**
- (iii) $(w + \bar{z})(\bar{w} + z)$ **2**
- (iv) If $\frac{w(9 - 8i)}{u}$ is a real number,
- (α) show that $a = 2b$, **2**
- (β) hence find u if $|u| = 5$. **2**

Question 2 (continued)

- (b) A, B are the points representing the complex numbers $z_1 = 13 + 11i$ and $z_2 = 10 + 8i$ respectively. Find the complex number that is represented by the point C where A, B and C are adjacent vertices of a regular hexagon.
- (c) (i) Express $z = \frac{1+i}{\sqrt{3}+i}$ in the form of $r \operatorname{cis} \theta$.
- (ii) Find the smallest positive integer n such that z^n is a purely imaginary number.



Marks

3

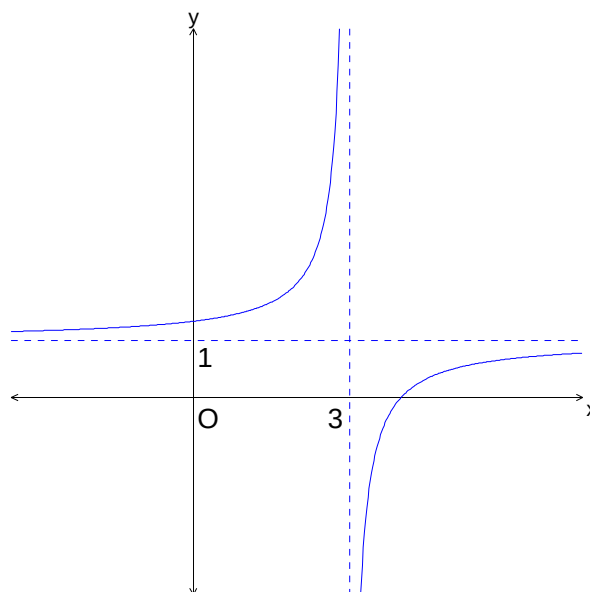
3

1

Question 3

(Begin a new page)

- (a) The diagram shows the graph of the function $f(x) = \frac{x-4}{x-3}$. On separate diagrams, sketch the following graphs, showing clearly any intercepts on the axes and the equations of the asymptotes.



- (i) $y = f(-x)$
- (ii) $y = |f(x)|$
- (iii) $y = f(|x|)$
- (iv) $y = e^{f(x)}$ and
- (v) $y = \sqrt{f(x)}$
- (b) The line $y = mx$ through the origin O (0, 0) touches the curve $y = \frac{x-4}{x-3}$ at point $P(x_1, y_1)$.
- (i) By considering the gradient of the line OP, show that $x_1^2 - 8x_1 + 12 = 0$

1

1

2

2

1

2

Question 3b continued on p.3...

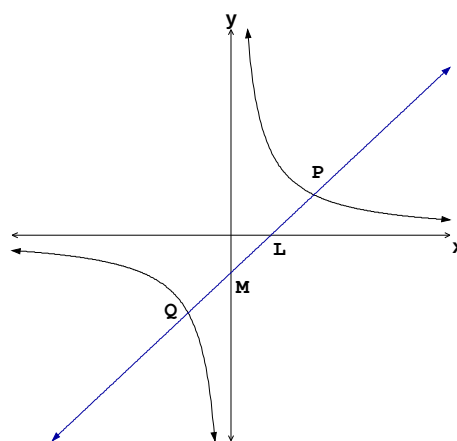
Question 3b (continued)

- (ii) Hence find the equation of the two tangents to $y = \frac{x-4}{x-3}$ from the origin. 2
- (c) (i) Sketch the region in the complex plane for which $|z - 5 - 12i| \leq 3$. 2
- (ii) Find the maximum modulus of the complex number z that satisfies $|z - 5 - 12i| \leq 3$. 2

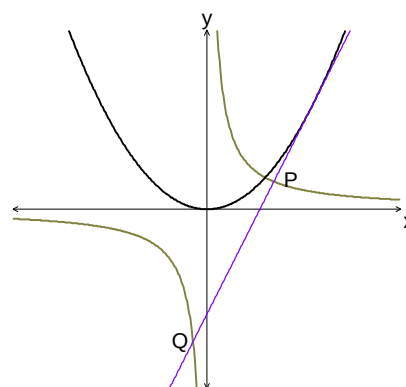
Question 4 (Begin a new page)

- (a) Given the equation of an ellipse is $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
- (i) Show that if the line $y = mx + c$ is a tangent to E , then $c^2 = a^2 m^2 + b^2$. 2
- (ii) Show that if the line $y = mx + c$ is a tangent to the circle $x^2 + y^2 = r^2$, then $c^2 = r^2 m^2 + r^2$. 1
- (iii) Hence or otherwise, find the equations of the common tangents to the ellipse $\frac{x^2}{6} + \frac{y^2}{3} = 1$ and the circle $2x^2 + 2y^2 = 9$. 3

- (b) $P \left(cp, \frac{c}{p} \right)$ and $Q \left(cq, \frac{c}{q} \right)$ are two points on the rectangular hyperbola $xy = c^2$. The chord PQ meets the asymptotes at L and M .



- (i) Show that the equation of the chord PQ is $x + pqy = c(p + q)$. 1
- (ii) Prove that $PL = QM$. 3
- (iii) A tangent to the parabola $x^2 = 4y$ meets the hyperbola $xy = 9$ at two points P and Q .
- (α) show that $1 + 3pq(p + q) = 0$. 2
- (β) Hence show that the equation of the locus of N , the mid-point of PQ , is $2x^2 + y = 0$. 3

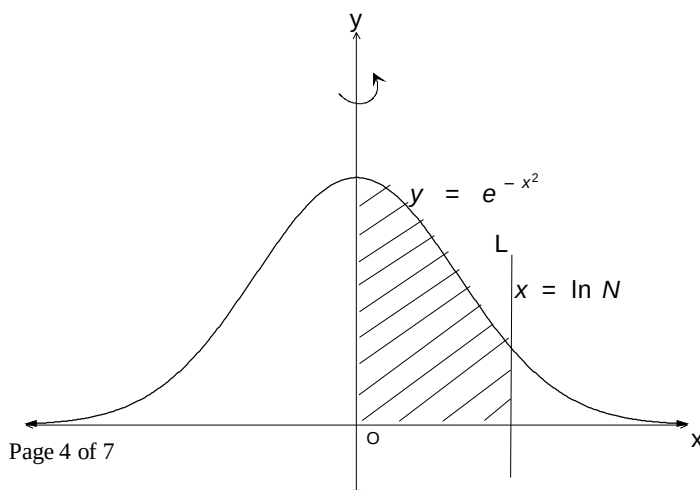


Question 5**(Begin a new page)****Marks**

- (a) The equation $x^3 - x^2 + 3x + 4 = 0$ has roots α , β and γ .
- (i) Find the values of $\alpha^2 + \beta^2 + \gamma^2$ and hence deduce that two of the roots must be complex. **3**
- (ii) Find the monic polynomial equation with roots α^2 , β^2 , and γ^2 . **2**
- (b) (i) Show that if $x = \alpha$ is a repeated root of the polynomial equation $P(x) = 0$, then it must also be a root of $P'(x) = 0$. **1**
- (ii) The hyperbola $xy = c^2$ touches the ellipse $\frac{(x-1)^2}{6} + y^2 = 1$ at point Q which lies in the 3rd quadrant. Deduce that the equation $x^4 - 2x^3 - 5x^2 + 6c^4 = 0$ has a negative repeated root α . **2**
- (iii) Hence determine the values of α and c^4 . **2**
- (c) (i) By using the identity $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$, show that the roots of the equation $16x^5 - 20x^3 + 5x - 1 = 0$ are $x = \cos \frac{2r\pi}{5}$, $r = 0, 1, 2, 3, 4$. **2**
- (ii) Hence show that $\cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4}$. **3**

Question 6**(Begin a new page)**

- (a) (i) Find $\int x e^{-x^2} dx$ using the substitution $u = x^2$. **2**
- (ii) The region bounded between the curve $y = e^{-x^2}$, the x-axis, the y-axis and the line L: $x = \log_e N$ is rotated about the y-axis. Show that the volume of the solid formed is **4**
- $\pi \left[1 - \left(\frac{1}{N} \right)^{\ln N} \right]$ by using the cylindrical shell method.

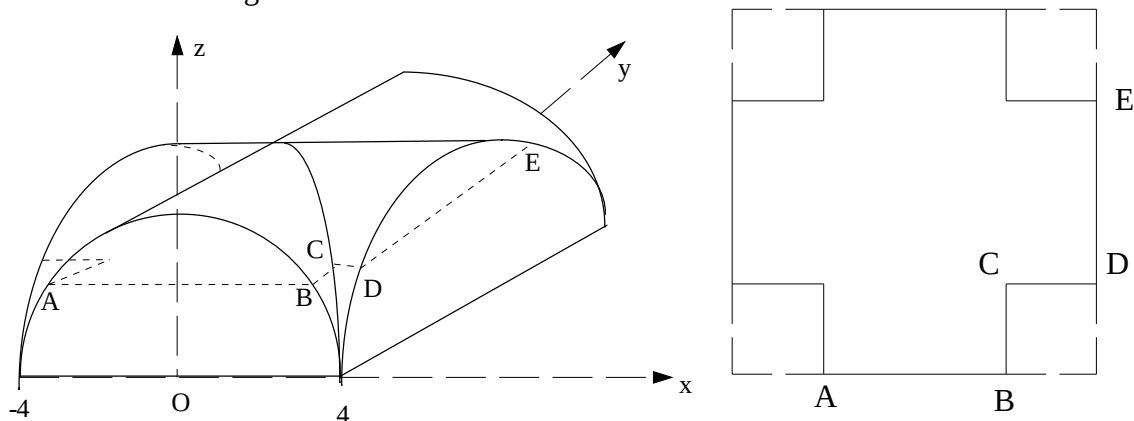


Question 6 (continued)

Marks

- (iii) Hence find the volume of the largest solid possible by sliding the line L indefinitely towards the right. 2

(b)



The solid shown above is formed in the shape of 2 identical intersecting semi-circular cylinders, each of radius 4. The base of the solid is a square of sides 8 cm. On the front face, the equation of the semi-circle is $x^2 + z^2 = 16$, $z \geq 0$, where x is the horizontal distance measured from the mid-point of the base of the front face and z is the height.

The shape of a horizontal slice of thickness δz taken at a height z is also shown above. It is a square with 4 smaller squares removed, one from each corner.

- (i) Find x in terms of z . 1

- (ii) Show that the volume of the solid is given by the expression 3

$$V = \int_0^4 \{8^2 - 4[4 - \sqrt{16 - z^2}]^2\} dz$$

- (iii) Hence find the volume of the solid. 3

Question 7

(Begin a new page)

- (a) The air resistance encountered by a body of mass m moving at a speed v is given by $kmgv$ where m is the mass of the body, g is the acceleration due to gravity and k is a constant.

- (i) Find the terminal velocity V_T . 2

- (ii) The body is projected vertically upward with an initial speed of U , show that the maximum height reached will be given by $H = \frac{U}{gk} - \frac{1}{gk^2} \ln(1 + kU)$. 3

Question 7a continued on p.6...

Question 7a (continued)**Marks**

- (iii) Show that, if the initial speed of projection is V_T , then the time taken to reach the maximum height is $T = \frac{V_T}{g} \ln 2$.

3

- (b) A body of mass m is projected vertically downward with a speed $\frac{1}{3} V_T$, where V_T is the terminal velocity, in a resistive medium. The resistance is mkv^2 , where k is a constant and v is the speed of the body t seconds after being projected. Show that it takes $\frac{V_T}{2g} \ln \frac{3}{2}$ seconds to reach a speed of $\frac{1}{2} V_T$.

4

- (c) The gravitational force exerted by the earth on a body of mass m at a distance x ($x \geq R$) from its centre is given by $\frac{mk}{x^2}$, where R is the radius of the earth and k is a constant.

- (i) Show that $k = gR^2$.

1

- (ii) A body is projected vertically upward with an initial speed U . Show that the speed of the body when it is at a distance x from the centre of the earth is given by

2

$$v^2 = U^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right).$$

Question 8**(Begin a new page)**

- (a) A sequence of number $\{u_n\}$ is defined by $u_1 = 1, u_2 = 5$ and

3

$$u_{n+2} = 5u_{n+1} - 6u_n \quad n \geq 1$$

Use mathematical induction to show that $u_n = 3^n - 2^n \quad n \geq 1$

- (b) Show that

3

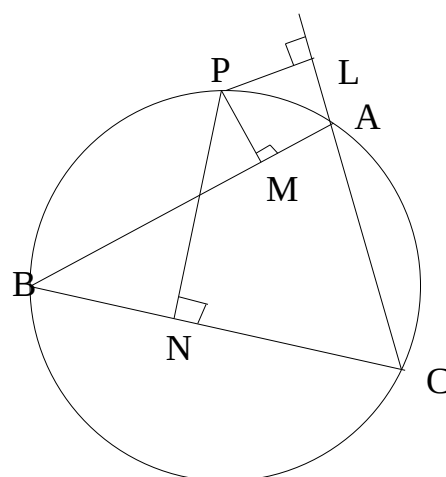
$$\sin \theta + \sin 2\theta + \dots + \sin n\theta = \frac{1}{2 \sin \frac{\theta}{2}} \left[\cos \frac{\theta}{2} - \cos \frac{(2n+1)\theta}{2} \right].$$

[You may assume the identity $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$]

Question 8 (continued)

- (c) P is a point on the minor arc AB. L, M and N are the feet of perpendiculars from P to CA, AB and BC respectively.

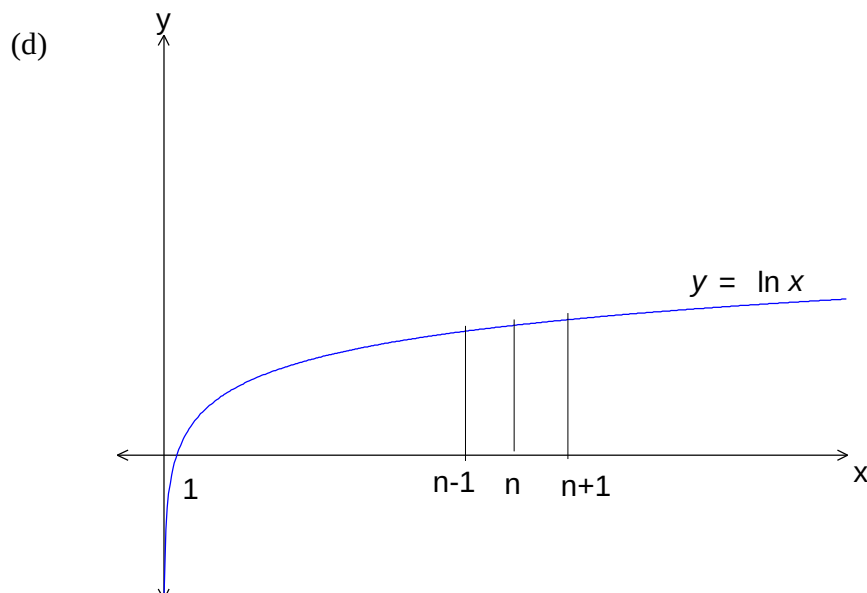
- (i) Copy the diagram.
- (ii) State the reason why P, M, A and L are concyclic.
- (iii) State the reason why P, B, N and M are concyclic.
- (iv) Show that L, M and N are collinear.



1

1

2



- (i) By considering the appropriate area under the graph of $y = \log_e x$, show that

2

$$\log_e n < \int_n^{n+1} \log_e x \, dx < \log_e (n+1) \quad n \geq 1$$

- (ii) Hence, show that $1 + \frac{1}{n} < e^{\frac{1}{n}} < \left(1 + \frac{1}{n}\right)^{\frac{n+1}{n}}$

3

End of Paper

Question 1

(a) (i) $\frac{1}{e^x - 1} = \frac{1}{e^x - 1} \times \frac{e^{-x}}{e^{-x}}$
 $= \frac{e^{-x}}{1 - e^{-x}}$ ✓

(ii) $\int \frac{1}{e^x - 1} dx = \int \frac{e^{-x}}{1 - e^{-x}} dx$
 $= \int \frac{d(1 - e^{-x})}{1 - e^{-x}}$
 $= \ln(1 - e^{-x})$ ✓

(b) $u = 1 + x$

$\therefore x = u - 1$

$dx = du$

$\int \frac{x}{\sqrt{1-x}} dx = \int \frac{u-1}{\sqrt{u}} du$

$= \int \sqrt{u} - \frac{1}{\sqrt{u}} du$ ✓

$= \frac{2}{3} u^{3/2} - 2\sqrt{u} + C$

$= \frac{2}{3} (1+x)^{3/2} - 2\sqrt{1+x} + C$ ✓

(d) (i) $\int \tan^{-1} x dx = x \tan^{-1} x - \int x d(\tan^{-1} x)$

$= x \tan^{-1} x - \int \frac{x}{1+x^2} dx$ ✓

$= x \tan^{-1} x - \frac{1}{2} \int \frac{d(1+x^2)}{1+x^2}$

$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$ ✓

(ii) Put $u = x^2$

$du = 2x dx$

$\therefore \int x \tan^{-1}(x^2) dx = \frac{1}{2} \int \tan^{-1} u du$ ✓

$= \frac{1}{2} u \tan^{-1} u - \frac{1}{4} \ln(1+u^2) + C$

$= \frac{x^2}{2} \tan^{-1}(x^2) - \frac{1}{4} \ln(1+x^4) + C$ ✓

(e) (i) $\int \frac{dx}{x^2 + 2x + 10} = \int \frac{dx}{(x+1)^2 + 9}$

$= \frac{1}{3} \tan^{-1} \frac{x+1}{3} + C$

(ii) $\int_0^\pi x \cos 2x dx = \int_0^\pi (\pi - x) \cos 2(\pi - x) dx$

$= \int_0^\pi (\pi - x) \cos 2x dx$

$= \pi \int_0^\pi \cos 2x dx - \int_0^\pi x \cos 2x dx$ ✓

$2 \int_0^\pi x \cos 2x dx = \pi \int_0^\pi \cos 2x dx$

(c) Let $\frac{x^2 + x - 4}{(2x+1)(x^2+4)} = \frac{A}{2x+1} + \frac{Bx+C}{x^2+4}$

$\therefore A(x^2+4) + (Bx+C)(2x+1) = x^2 + x - 4$ (1)

Put $x = -\frac{1}{2}$

$A(\frac{1}{4} + 4) = \frac{1}{4} - \frac{1}{2} - 4$

$\frac{17}{4} A = -\frac{17}{4}$

$A = -1$

Equating coeff. of x^2 in (1)

$A + 2B = 1$

$2B = 1 - A$

$2B = 2$

$B = 1$

Put $x = 0$ in (1)

$4A + C = -4$

$C = -4 - 4A$

$= -4 + 4$

$= 0$

$\therefore \frac{x^2 + x - 4}{(2x+1)(x^2+4)} = -\frac{1}{2x+1} + \frac{x}{x^2+4}$ ✓

$\therefore \int \frac{x^2 + x - 4}{(2x+1)(x^2+4)} dx = -\int \frac{1}{2x+1} dx + \int \frac{x}{x^2+4} dx$ ✓

$= -\frac{1}{2} \ln(2x+1) + \frac{1}{2} \ln(x^2+4) + C$

$= \frac{1}{2} \ln \frac{x^2+4}{2x+1} + C$

P.1

$\therefore \int_0^\pi x \cos 2x dx = \frac{1}{2} \pi \int_0^\pi \cos 2x dx$

$= \frac{1}{4} \pi [\sin 2x]_0^\pi$

$= 0$ ✓

Question 2

(a) (i) $w + \bar{z} = (2+5i) + (4-3i)$

$= 2+5i + 4-3i$

$= 6+2i$ ✓

(ii) $|w + \bar{z}| = \sqrt{6^2 + 2^2}$

$= 10$ ✓

(iii) $(w + \bar{z})(\bar{w} + z) = (w + \bar{z})(\overline{w + \bar{z}})$

$= |w + \bar{z}|^2$ ✓

$= 10^2$

$= 100$ ✓

(iv) (a) $\frac{w(9-8i)}{u} = \frac{(2+5i)(9-8i)}{a+bi}$

$= \frac{(18+40) + (45-16)i}{a+bi}$

$= \frac{58+29i}{a+bi} \times \frac{a-bi}{a-bi}$

$= \frac{(58a+29b) + (29a-58b)i}{a^2-b^2}$ ✓

P.2

$$\frac{w(9-8i)}{u} \text{ is real only if } \operatorname{Im}\left(\frac{w(9-8i)}{u}\right) = 0$$

$$\text{i.e. } \frac{29a-58b}{a^2+b^2} = 0$$

$$\text{i.e. } a = 2b$$

$$(p) \quad u = a + bi$$

$$\therefore |u| = \sqrt{a^2 + b^2}$$

$$|u| = 5$$

$$a^2 + b^2 = 25$$

$$(2b)^2 + b^2 = 25$$

$$5b^2 = 25$$

$$b^2 = 5$$

$$b = \pm\sqrt{5}$$

$$\therefore u = \pm\sqrt{5}(2+i)$$

$$(b) \quad \vec{BA} = \vec{z}_1 - \vec{z}_2$$

$$= (13+11i) - (10+8i)$$

$$= 3+3i$$

$$\vec{BC} = \vec{BA} \times \operatorname{cis} \frac{2\pi}{3}$$

$$= (3+3i) \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$= (3+3i) \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$\text{i.e. } \frac{n\pi}{12} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots$$

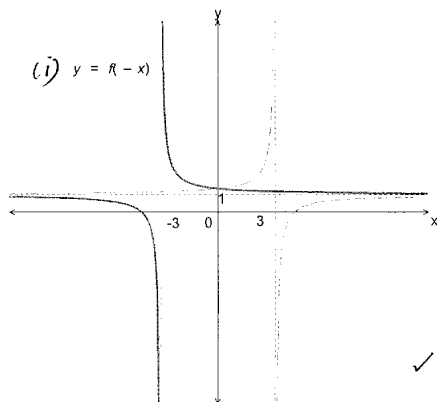
$$n = 6, 18, 30, 42, \dots$$

$\therefore$ The smallest positive integral value of $n = 6$.

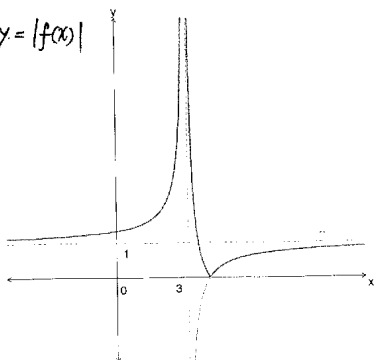
Question 3

(a)

$$(i) \quad y = f(x-x)$$



$$(ii) \quad y = |f(x)|$$



$$= -\frac{3}{2} - \frac{3\sqrt{3}}{2} + \left(\frac{3\sqrt{3}}{2} - \frac{3}{2} \right) i$$

$$\text{but } \vec{BC} = \vec{z}_3 - \vec{z}_2 \quad \text{where } \vec{z}_3 \text{ is the complex no. that is represented by C.}$$

$$\therefore \vec{z}_3 = \vec{z}_2 + \vec{BC}$$

$$= 10+8i - \frac{3}{2} - \frac{3\sqrt{3}}{2} + \left(\frac{3\sqrt{3}}{2} - \frac{3}{2} \right) i$$

$$= \left(\frac{17}{2} - \frac{3\sqrt{3}}{2} \right) + \left(\frac{13}{2} + \frac{3\sqrt{3}}{2} \right) i$$

$$= \frac{17-3\sqrt{3}}{2} + \frac{13+3\sqrt{3}}{2} i$$

$$(c) (i) \quad \vec{z} = \frac{1+i}{\sqrt{3}+i}$$

$$= \frac{\sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right)}{2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2} i \right)}$$

$$= \frac{1}{\sqrt{2}} \frac{\operatorname{cis} \frac{\pi}{4}}{\operatorname{cis} \frac{\pi}{6}}$$

$$= \frac{1}{\sqrt{2}} \operatorname{cis} \left(\frac{\pi}{4} - \frac{\pi}{6} \right)$$

$$= \frac{1}{\sqrt{2}} \operatorname{cis} \frac{\pi}{12}$$

$$(ii) \quad \vec{z}^n = \left(\frac{1}{\sqrt{2}} \right)^n \operatorname{cis} \frac{n\pi}{12}$$

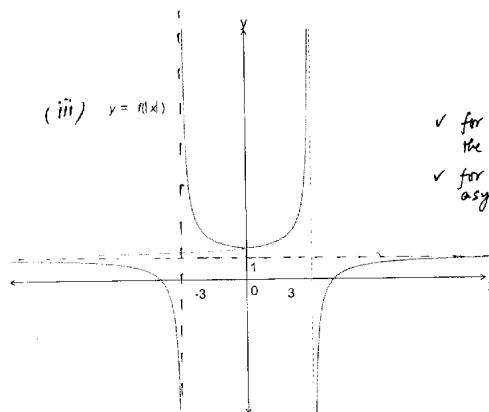
$$= \left(\frac{1}{\sqrt{2}} \right)^n \left(\cos \frac{n\pi}{12} + i \sin \frac{n\pi}{12} \right)$$

if $\vec{z}^n$ is a purely imaginary number,

$$\text{then } \cos \frac{n\pi}{12} = 0$$

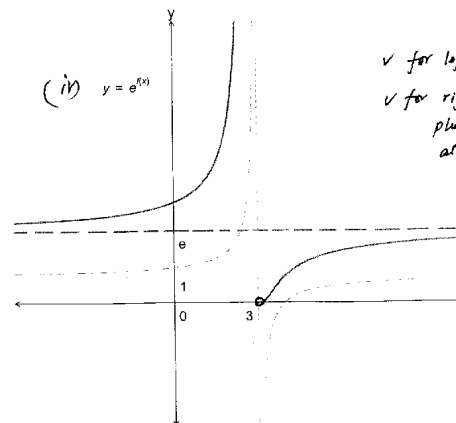
p. 3

$$(iii) \quad y = f(x)$$



✓ for reflection about the y-axis
✓ for all correct asymptotes

$$(iv) \quad y = e^{f(x)}$$

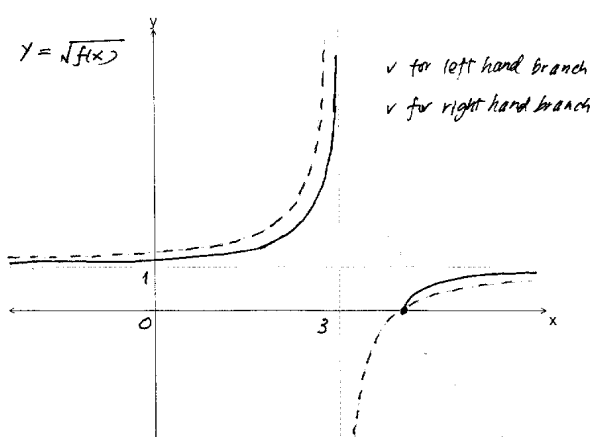


✓ for left hand branch
✓ for right hand branch plus open end-pt at $x=3$.

p. 4

3(a)

$$(v) y = \sqrt{x-3}$$



$$(b) (i) m_{op} = \frac{y_1}{x_1}$$

(1)

$$y = \frac{x-4}{x-3}$$

$$y' = \frac{(x-3) - (x-4)}{(x-3)^2}$$

$$= \frac{1}{(x-3)^2}$$

$$\therefore \text{Gradient at } P(x_1, y_1) \text{ is } \frac{1}{(x_1-3)^2} \quad (2)$$

$$\text{From (1) \& (2)} \quad \frac{y_1}{x_1} = \frac{1}{(x_1-3)^2}$$

$$\text{but } y_1 = \frac{x_1-4}{x_1-3}$$

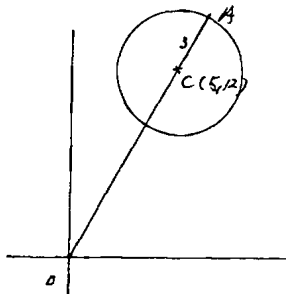
(ii) OCA is a straight line where C is the centre of the circle

$$OC = \sqrt{5^2 + 12^2}$$

$$= 13$$

$$\therefore OA = 13 + 3$$

$$= 16$$



Question 4

$$(a) (i) y = mx + c$$

(1)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(2)

Put (1) into (2)

$$\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$b^2x^2 + a^2(mx+c)^2 = a^2b^2$$

$$b^2x^2 + a^2(m^2x^2 + 2mcx + c^2) = a^2b^2$$

$$b^2x^2 + a^2m^2x^2 + 2a^2mcx + a^2c^2 - a^2b^2 = 0$$

$$(b^2 + a^2m^2)x^2 + 2a^2mcx + (a^2c^2 - a^2b^2) = 0 \quad (3) \quad \checkmark$$

If (1) is a tangent to E, then $\Delta = 0$

$$(2a^2mc)^2 - 4(b^2 + a^2m^2)(a^2c^2 - a^2b^2) = 0$$

$$4a^4m^2c^2 - 4a^2(b^2 + a^2m^2)(c^2 - b^2) = 0$$

$$a^2m^2c^2 - b^2c^2 + b^4 - a^2mc^2 + a^2b^2m^2 = 0$$

$$\therefore \frac{x_1-4}{x_1-3} = \frac{1}{(x_1-3)^2}$$

$$(x_1-3)(x_1-4) = x_1$$

$$x_1^2 - 7x_1 + 12 = x_1$$

$$x_1^2 - 8x_1 + 12 = 0$$

$$(ii) x_1^2 - 8x_1 + 12 = 0$$

$$(x_1-2)(x_1-6) = 0$$

$$\therefore x_1 = 2 \text{ or } x_1 = 6$$

$$\text{When } x_1 = 2, y_1 = \frac{2-4}{2-3} = 2$$

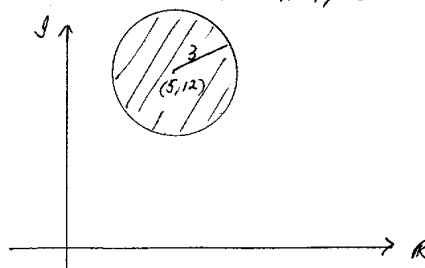
$$\therefore \text{Equation of tangent is } y = x$$

$$\text{When } x_1 = 6, y_1 = \frac{6-4}{6-3} = \frac{2}{3}$$

$$\therefore \text{Equation of tangent is } y = \frac{x}{3}$$

$$\text{or } x - 3y = 0$$

(c) (i)



1 for correct region

1 for correct centre & radius.

P.5.

$$b^4 + a^2b^2m^2 - b^2c^2 = 0$$

$$b^2 + a^2m^2 - c^2 = 0$$

$$\therefore b \neq 0$$

$$\therefore c^2 = a^2m^2 + b^2$$

$$(ii) \text{Substitute } c^2 = b^2 + r^2 \text{ into } c^2 = a^2m^2 + b^2$$

$$\therefore c^2 = r^2m^2 + r^2$$

$$(iii) \text{If } y = mx + c \text{ is a common tangent,}$$

$$\text{then } c^2 = a^2m^2 + b^2 = r^2m^2 + r^2$$

$$\text{ie } 6m^2 + 3 = \frac{9}{2}m^2 + \frac{9}{2}$$

$$12m^2 + 6 = 9m^2 + 9$$

$$3m^2 = 3$$

$$m = \pm 1$$

$$\text{When } m = 1, c^2 = 6 + 3$$

$$= 9$$

$$c = \pm 3$$

$\therefore$ The equations of the common tangents are

$$y = x \pm 3$$

$$\text{When } m = -1, c^2 = 6 + 3$$

$$c = \pm 3$$

$\therefore$ The equations are $y = -x \pm 3$.

P.6

(b) (i) $P(cp, \frac{c}{p}), Q(cq, \frac{c}{q})$

∴ Equation of chord PQ is

$$\frac{y - \frac{c}{q}}{x - cq} = \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq}$$

$$y - \frac{c}{q} = \frac{\frac{1}{p} - \frac{1}{q}}{p - q} (x - cq)$$

$$pq(y - \frac{c}{q}) = pq \frac{\frac{1}{p} - \frac{1}{q}}{p - q} (x - cq)$$

$$pqy - cp = \frac{q - p}{p - q} (x - cq)$$

$$pqy - cp = cq - x$$

$$\text{ie } x + pqy = c(p+q) \quad (1) \quad \checkmark$$

(ii) At L, $y=0$

$$\therefore x = c(p+q)$$

$$\therefore L \text{ is } (c(p+q), 0)$$

At M, $x=0$,

$$\text{from (1)} \quad pqy = c(p+q)$$

$$\therefore y = \frac{c(p+q)}{pq}$$

$$\therefore M \text{ is } (0, \frac{c(p+q)}{pq})$$

$$x + \frac{pqx^2}{4} = 3(p+q)$$

$$4x + pqx^2 = 12(p+q)$$

$$pqx^2 + 4x - 12(p+q) = 0 \quad \checkmark$$

Since PQ is a tangent to $x^2 = 4y$

$$\therefore \Delta = 0$$

$$\text{ie } 4^2 + 4pq(12)(p+q) = 0$$

$$16 + 48pq(p+q) = 0$$

$$1 + 3pq(p+q) = 0 \quad (3) \quad \checkmark$$

(β) At N,

$$x = \frac{3}{2}(p+q) \quad (4)$$

$$y = \frac{3}{2}(\frac{1}{p} + \frac{1}{q})$$

$$= \frac{3(p+q)}{2pq} \quad (5)$$

$$\frac{(4)}{(5)} \quad pq = \frac{x}{y} \quad (6) \quad \checkmark$$

$$\text{From (3)} \quad p+q = \frac{2x}{3} \quad (7) \quad \checkmark$$

Put (6) & (7) into (3)

$$1 + 3(\frac{x}{y})(\frac{2x}{3}) = 0 \quad \checkmark$$

$$1 + \frac{2x^2}{y} = 0$$

mid-pt of PQ is

$$N(\frac{c(p+q)}{2}, \frac{c}{2}(\frac{1}{p} + \frac{1}{q}))$$

$$= (\frac{c(p+q)}{2}, \frac{c}{2} \cdot \frac{p+q}{pq})$$

$$= (\frac{c(p+q)}{2}, \frac{c(p+q)}{2pq})$$

mid-pt of LM is

$$N' = (\frac{c(p+q)}{2}, \frac{c(p+q)}{2pq})$$

$$\therefore N = N' \quad \checkmark$$

Hence $PN = QN$

$$LN = MN$$

$$PN - LN = QN - MN$$

$$\therefore PL = QM. \quad \checkmark$$

(iii) (a) $xy = 9 \quad c=3$

∴ Let P & Q be the pts $(3p, \frac{3}{p}) + (3q, \frac{3}{q})$ respectively.

By result of (i) eqⁿ of PQ is

$$x + pqy = 3(p+q) \quad (2)$$

When this line meets the parabola $x^2 = 4y$,

$$\text{put } y = \frac{x^2}{4} \text{ into (2)}$$

P.7

$$\text{ie } 2x^2 + y = 0$$

$$\therefore \text{Equation of the locus of N is } 2x^2 + y = 0$$

Question 5

$$\begin{aligned} (a) \text{ (i) } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 1 - 2(3) \\ &= -5 \end{aligned} \quad \checkmark$$

Since sum of squares of real numbers must be non-negative, hence not all α, β, γ are real, but complex roots of a polynomial equation occur in conjugate pairs, ∴ exactly 2 of α, β, γ are complex. $\checkmark$

(ii) Let $y = x^2$

$$\therefore x^3 - x^2 + 3x + 4 = 0 \text{ is transformed into}$$

$$y\sqrt{y} - y + 3\sqrt{y} + 4 = 0$$

$$\sqrt{y}(y+3) = y-4 \quad \checkmark$$

$$y(y+3)^2 = (y-4)^2$$

$$y(y^2 + 6y + 9) = y^2 - 8y + 16$$

$$y^3 + 6y^2 + 9y = y^2 - 8y + 16$$

$$y^3 + 5y^2 + 17y - 16 = 0$$

∴ Equation whose roots are $\alpha^2, \beta^2, \gamma^2$ is

$$x^3 + 5x^2 + 17x - 16 = 0 \quad \checkmark \text{ P.8}$$

(b) i) Let $P(x) = (x-\alpha)^2 Q(x)$

$$P'(x) = 2(x-\alpha)Q(x) + (x-\alpha)^2 Q'(x)$$

$$= (x-\alpha)[2Q(x) + (x-\alpha)Q'(x)]$$

$$\therefore P'(\alpha) = 0$$

hence $x = \alpha$ must also be a root of $P'(x) = 0$

(ii) When $xy = c^2$ meets $\frac{(x-1)^2}{6} + y^2 = 1$

$$\frac{(x-1)^2}{6} + \left(\frac{c^2}{x}\right)^2 = 1$$

$$x^2(x-1)^2 + 6c^4 = 6x^2$$

$$x^2(x^2 - 2x + 1) + 6c^4 = 6x^2$$

$$x^4 - 2x^3 + x^2 + 6c^4 - 6x^2 = 0$$

$$x^4 - 2x^3 - 5x^2 + 6c^4 = 0 \quad (1) \quad \checkmark$$

Since $xy = c^2$ touches $\frac{(x-1)^2}{6} + y^2 = 1$ at Q

$\therefore$ equation (1) must have a repeated root.

Q lies in the 3rd quadrant, $\therefore x$ -coordinate of Q must be negative

$$\therefore \alpha < 0$$

$\therefore$ (1) must have a negative repeated root α

(ii) Product of roots of $16x^5 - 20x^3 + 5x - 1 = 0$

$$\therefore \cos 0 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} \cos \frac{6\pi}{5} \cos \frac{8\pi}{5} = \frac{1}{16} \quad \checkmark$$

$$\cos \frac{2\pi}{5} \cos(\pi - \frac{\pi}{5}) \cos(\pi + \frac{\pi}{5}) \cos(2\pi - \frac{2\pi}{5}) = \frac{1}{16}$$

$$\cos \frac{2\pi}{5} (-\cos \frac{\pi}{5}) (-\cos \frac{\pi}{5}) \cos \frac{2\pi}{5} = \frac{1}{16}$$

$$\left(\cos \frac{\pi}{5} \cos \frac{2\pi}{5}\right)^2 = \frac{1}{16}$$

$$\cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4} \quad \checkmark$$

Since $\cos \frac{\pi}{5} > 0$, $\cos \frac{2\pi}{5} > 0 \quad \checkmark$

Question 6

(a) i) $u = x^2$

$$du = 2x dx$$

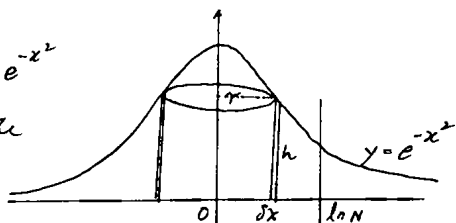
$$\therefore \int x e^{-x^2} dx = \frac{1}{2} \int e^{-u} du \quad \checkmark$$

$$= -\frac{1}{2} e^{-u} + C$$

$$= -\frac{1}{2} e^{-x^2} + C \quad \checkmark$$

(ii) $r = x$,
 $h = y = e^{-x^2}$

Volume of the cylindrical shell is



(iii) Let $P(x) = x^4 - 2x^3 - 5x^2 + 6x^4$

$$P'(x) = 4x^3 - 6x^2 - 10x$$

$$P'(x) = 0$$

$$2x(2x^2 - 3x - 5) = 0$$

$$2x(2x - 5)(x + 1) = 0$$

$$\therefore x = 0, \frac{5}{2} \text{ or } -1.$$

Since $\alpha < 0$

$$\therefore \alpha = -1 \quad \checkmark$$

$$P(-1) = 0$$

$$(-1)^4 - 2(-1)^3 - 5(-1)^2 + 6(-1)^4 = 0$$

$$1 + 2 - 5 + 6 = 0$$

$$6 = 5$$

$$\therefore 6 = 5 \quad \checkmark$$

(c) i) $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$

Put $x = \cos \theta$

Roots of $\cos 5\theta = 1 = \cos 0$

are $\cos \theta = \cos \frac{2r\pi}{5} \quad r = 0, 1, 2, 3, 4 \quad \checkmark$

$\therefore$ Roots of $16x^5 - 20x^3 + 5x - 1 = 0$

ie $16x^5 - 20x^3 + 5x - 1 = 0$

are $x = \cos \frac{2r\pi}{5} \quad r = 0, 1, 2, 3, 4 \quad \checkmark \quad \text{p. 9}$

$$\delta V = 2\pi r h \delta x$$

$$= 2\pi x e^{-x^2} \delta x$$

$$\therefore \text{Volume} = \lim_{N \rightarrow \infty} \sum \delta V$$

$$= \lim_{N \rightarrow \infty} \sum 2\pi x e^{-x^2} \delta x$$

$$= 2\pi \int_0^{\ln N} x e^{-x^2} dx \quad \checkmark$$

$$= 2\pi \left[-\frac{1}{2} e^{-x^2} \right]_0^{\ln N} \quad (\text{from (i)})$$

$$= \pi (1 - e^{-(\ln N)^2}) \quad \checkmark$$

$$= \pi [1 - e^{-(\ln N)^2}]$$

$$= \pi \left[1 - \left(\frac{1}{N} \right)^{\ln N} \right] \quad \checkmark$$

(iii) Let $u = \left(\frac{1}{N} \right)^{\ln N}$

$$\therefore \ln u = \ln N \ln \left(\frac{1}{N} \right)$$

$$= -(\ln N)^2$$

$\therefore$ as $N \rightarrow \infty$ $\ln u$ will decrease without bound, ie $u \rightarrow 0 \quad \checkmark$

$$\therefore \lim_{N \rightarrow \infty} \pi \left[1 - \left(\frac{1}{N} \right)^{\ln N} \right] = \pi (1 - 0) = \pi \quad \checkmark$$

(b) (i) $x^2 + z^2 = 16$
 $x = 16 - z^2$
 $x = \sqrt{16 - z^2}$ ✓

(ii) Each side of the square = δ

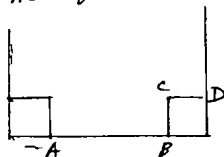
$$AB = 2x$$

$$= 2\sqrt{16 - z^2}$$

∴ Length of each side of the smaller squares at the corners

$$= \frac{1}{2} [\delta - 2\sqrt{16 - z^2}]$$

$$= 4 - \sqrt{16 - z^2}$$
 ✓



∴ Sum of areas of the squares at the corners

$$= 4 [4 - \sqrt{16 - z^2}]^2$$

$$\therefore \text{Area of the slice} = \delta^2 - 4 [4 - \sqrt{16 - z^2}]^2$$
 ✓

$$\therefore \text{Volume of the slice } \delta V = \{ \delta^2 - 4 [4 - \sqrt{16 - z^2}]^2 \} \delta z$$
 ✓

∴ Volume of the solid

$$V = \lim_{\delta z \rightarrow 0} \sum \delta V$$

$$= \int_0^4 \{ \delta^2 - 4 [4 - \sqrt{16 - z^2}]^2 \} dz$$

$$v \frac{dv}{dx} = -g(1 + kv)$$

$$\int_u^0 \frac{v dv}{1 + kv} = - \int_0^H g dx$$
 ✓

$$\frac{1}{k} \int_u^0 (1 - \frac{1}{1 + kv}) dv = -g[x]_0^H$$

$$\frac{1}{k} [v - \frac{1}{k} \ln(1 + kv)]_u^0 = -gH$$
 ✓

$$\frac{1}{k} [(0 - \frac{1}{k} \ln 1) - u + \frac{1}{k} \ln(1 + ku)] = -gH$$

$$\therefore H = \frac{1}{gk} [u - \frac{1}{k} \ln(1 + ku)]$$

$$= \frac{u}{gk} - \frac{1}{gk^2} \ln(1 + ku)$$

(iii) For upward motion:

$$\ddot{x} = -g(1 + kv)$$

$$\frac{dv}{dt} = -g(1 + kv)$$

$$\int_{v_T}^0 \frac{dv}{1 + kv} = - \int_0^T g dt$$
 ✓

$$\frac{1}{k} [\ln(1 + kv)]_{v_T}^0 = -[gt]_0^T$$

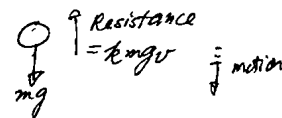
(iii) $V = \int_0^4 (64 - 16 + 4\sqrt{16 - z^2}) dz$
 $= \int_0^4 (48 + 4\sqrt{16 - z^2}) dz$ ✓
 $= [48z]_0^4 + 2 \times \text{area of semi-circle of radius 4}$
 $= 192 + 2 \times \frac{1}{2} \pi (4^2)$ ✓
 $= 192 + 16\pi$ ✓

Question 7

(a) (i) When the body is moving downward, take downward as positive direction.

$$m\ddot{x} = mg - kmgv$$

$$= mg(1 - kv)$$
 ✓



at terminal velocity, $\ddot{x} = 0$

$$\therefore 1 - kv_T = 0$$

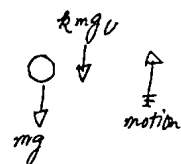
$$v_T = \frac{1}{k}$$
 ✓

(ii) Take upward as positive direction.

$$m\ddot{x} = -mg + kmgv$$

$$= -mg(1 + kv)$$

$$\ddot{x} = -g(1 + kv)$$
 ✓



P.11

$$\frac{1}{k} \ln(1 + kv_T) = gT$$

$$\therefore T = \frac{1}{gk} \ln(1 + kv_T)$$
 ✓

$$\text{but } v_T = \frac{1}{k}$$

$$\therefore T = \frac{v_T}{g} \ln(1 + k \cdot \frac{1}{k})$$

$$\therefore T = \frac{v_T}{g} \ln 2$$
 ✓

(b) $m\ddot{x} = mg - mkv^2$

$$\ddot{x} = g - kv^2$$

$$\frac{dv}{dt} = g - kv^2$$

$$\int_{\frac{v_T}{3}}^{\frac{v_T}{2}} \frac{dv}{g - kv^2} = \int_0^T dt$$
 ✓

$$\frac{1}{k} \int_{\frac{v_T}{3}}^{\frac{v_T}{2}} \frac{dv}{\frac{g}{k} - v^2} = T$$

$$T = \frac{1}{k} \int_{\frac{v_T}{3}}^{\frac{v_T}{2}} \frac{dv}{(\frac{\sqrt{g}}{k} - v)(\frac{\sqrt{g}}{k} + v)}$$

when terminal velocity is attained
 $\ddot{x} = 0$

P.12

$$g - kV_T^2 = 0$$

$$\therefore V_T = \sqrt{\frac{g}{k}} \quad \text{or } k = \frac{g}{V_T^2}$$

$$\therefore T = \frac{1}{k} \int_{\frac{V_T}{3}}^{\frac{V_T}{2}} \frac{dv}{(V_T - v)(V_T + v)} \quad \checkmark$$

$$= \frac{1}{2kV_T} \int_{\frac{V_T}{3}}^{\frac{V_T}{2}} \left(\frac{1}{V_T - v} + \frac{1}{V_T + v} \right) dv$$

$$= \frac{1}{2kV_T} \left[\ln(V_T + v) - \ln(V_T - v) \right]_{\frac{V_T}{3}}^{\frac{V_T}{2}}$$

$$= \frac{1}{2kV_T} \left[\ln \left(\frac{V_T + v}{V_T - v} \right) \right]_{\frac{V_T}{3}}^{\frac{V_T}{2}}$$

$$= \frac{1}{2kV_T} \left[\ln \left(\frac{V_T + \frac{V_T}{2}}{V_T - \frac{V_T}{2}} \right) - \ln \left(\frac{V_T + \frac{V_T}{3}}{V_T - \frac{V_T}{3}} \right) \right] \quad \checkmark$$

$$= \frac{1}{2kV_T} [\ln 3 - \ln 2]$$

$$= \frac{1}{2V_T} \cdot \frac{V_T^2}{g} \ln \frac{3}{2} \quad \therefore k = \frac{g}{V_T^2}$$

$$= \frac{V_T}{2g} \ln \frac{3}{2} \quad \checkmark$$

Question 8

$$(a) \quad U_n = 3^n - 2^n$$

$$\text{when } n=1, \quad U_1 = 3 - 2 = 1$$

$$\text{when } n=2 \quad U_2 = 3^2 - 2^2 = 5$$

$\therefore$ It is true for $n=1$ & $n=2$. $\checkmark$

Assume it is true for $n=k$ and $n=k+1$

$$\text{ie } U_k = 3^k - 2^k$$

$$U_{k+1} = 3^{k+1} - 2^{k+1} \quad \checkmark$$

$$\text{then } U_{k+2} = 5U_{k+1} - 6U_k$$

$$= 5(3^{k+1} - 2^{k+1}) - 6(3^k - 2^k) \quad \checkmark$$

$$= 15(3^k) - 10(2^k) - 6(3^k) + 6(2^k)$$

$$= 9(3^k) - 4(2^k)$$

$$= 3^2 \cdot 3^k - 2^2 \cdot 2^k$$

$$= 3^{k+2} - 2^{k+2} \quad \checkmark$$

Hence it will be true for $n=k+2$ if it is true for $n=k$ and $n=k+1$. Since it is proved true for $n=1$ and $n=2$,

$$(c) (i) \quad m\ddot{x} = -\frac{mk}{x^2}$$

at surface of earth, $x=R$ & $m\ddot{x} = -mg$

$$\therefore -mg = -\frac{mk}{R^2} \quad \checkmark$$

$$\therefore k = gR^2$$

$$(ii) \quad m\ddot{x} = -\frac{mk}{x^2}$$

$$\ddot{x} = -\frac{k}{x^2}$$

$$= -\frac{gR^2}{x^2}$$

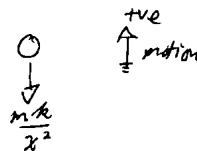
$$\frac{d}{dx} \left(\frac{v^2}{2} \right) = -\frac{gR^2}{x^2} \quad \checkmark$$

$$\therefore \int_u^v d \left(\frac{v^2}{2} \right) = \int_R^x \frac{gR^2}{x^2} dx$$

$$\left[\frac{v^2}{2} \right]_u^v = \left[\frac{gR^2}{x} \right]_R^x$$

$$\frac{V^2}{2} - \frac{U^2}{2} = gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$$

$$\therefore V^2 = U^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right) \quad \checkmark$$



P.13

$\therefore$ it will be true for $n=3, 4, 5, \dots$ i.e. all positive integers n .

$$(b) \quad \text{Let } S = \sin \theta + \sin 2\theta + \dots + \sin(n-1)\theta + \sin n\theta$$

$$\therefore 2S \cdot \sin \frac{\theta}{2} = 2\sin \theta \sin \frac{\theta}{2} + 2\sin 2\theta \sin \frac{\theta}{2} + \dots + 2\sin(n-1)\theta \sin \frac{\theta}{2} \quad \checkmark$$

$$+ 2\sin n\theta \sin \frac{\theta}{2}$$

$$= [\cos \frac{\theta}{2} - \cos \frac{3\theta}{2}] + [\cos \frac{3\theta}{2} - \cos \frac{5\theta}{2}] + \dots$$

$$+ [\cos \frac{(n-1)\theta}{2} - \cos \frac{(2n-1)\theta}{2}] + [\cos \frac{(2n-1)\theta}{2} - \cos \frac{(2n+1)\theta}{2}]$$

$$= \cos \frac{\theta}{2} - \cos \frac{(2n+1)\theta}{2} \quad \checkmark$$

$$\therefore S = \frac{1}{2\sin \frac{\theta}{2}} [\cos \frac{\theta}{2} - \cos \frac{(2n+1)\theta}{2}]$$

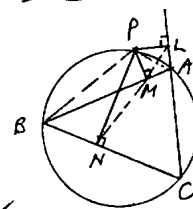
$$(c) (ii) \quad \angle PMA + \angle PLA = 2 \text{ rts}$$

$\therefore P, M, A, L$ are concyclic since opposite angles are supplementary $\checkmark$

$$(ii) \quad \angle PMB = \angle PNB$$

$\therefore P, B, M, N$ are concyclic since angles on the same chord in the same segment are equal $\checkmark$

$$(iii) \quad \begin{aligned} \angle PAL &= \angle PBC & (\text{ext } \angle \text{ of cyclic quad}) \\ \angle AML &= \angle APL & (\text{angles in the same segment}) \\ &= 90^\circ - \angle PAL & (\angle \text{ s sum of } \Delta) \end{aligned} \quad \checkmark$$



P.14

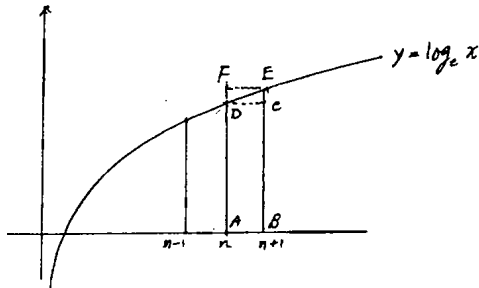
$$\therefore \angle AML = 90^\circ - \angle PBC$$

$$= \angle BPN \quad (\angle \text{sum of } \triangle PBN)$$

$$= \angle BMN \quad (\angle \text{'s in the same segment})$$

$\therefore L, M, N$ must be collinear (vert. opp $\angle$'s equal)

(d) (ii)



$$AD = \log_e n$$

$$AB = 1$$

$$\therefore \text{Area of } ABCD = \log_e n$$

Area of $ABCD < \text{area under } y = \log_e x \text{ between } x = n \text{ and } x = n+1$

$$\therefore \log_e n < \int_n^{n+1} \log_e x \, dx$$

$$\text{Similarly } \int_n^{n+1} \log_e x \, dx < \text{area of } ABEF$$

$$\int_n^{n+1} \log_e x \, dx < \log_e(n+1)$$

$$\therefore \log_e n < \int_n^{n+1} \log_e x \, dx < \log_e(n+1)$$

$$(ii) \log_e n < \left[x \log_e x - x \right]_n^{n+1} < \log_e(n+1)$$

$$\log_e n < (n+1) \log_e(n+1) - (n+1) - n \log_e n + n < \log_e(n+1)$$

$$\log_e n < (n+1) \log_e(n+1) - n \log_e n - 1 < \log_e(n+1) \quad (1)$$

By considering the LHS & middle section

$$(n+1) \log_e n < (n+1) \log_e(n+1) - 1$$

$$1 < (n+1) \log_e \left(\frac{n+1}{n} \right)$$

$$\frac{1}{n} < \left(\frac{n+1}{n} \right) \log_e \left(\frac{n+1}{n} \right)$$

$$e^{\frac{1}{n}} < \left(\frac{n+1}{n} \right)^{\frac{n+1}{n}}$$

$$e^{\frac{1}{n}} < \left(1 + \frac{1}{n} \right)^{\frac{n+1}{n}}$$

(2) ✓

Excluding the left-hand most side of (1)

$$(n+1) \log_e(n+1) - n \log_e n - 1 < \log_e(n+1)$$

$$n \log_e(n+1) - n \log_e n < 1$$

$$n \log_e \left(\frac{n+1}{n} \right) < 1$$

$$\log_e \left(\frac{n+1}{n} \right) < \frac{1}{n}$$

$$\therefore 1 + \frac{1}{n} < e^{\frac{1}{n}}$$

(3) ✓

From (2) & (3)

$$1 + \frac{1}{n} < e^{\frac{1}{n}} < \left(1 + \frac{1}{n} \right)^{\frac{n+1}{n}}$$

P15